Vansh Gupta

+91 9289011325 | vanshgupta7266@gmail.com | LinkedIn | GitHub
CodeChef 4-Star (max 1856) | Codeforces Pupil (max 1340) | Bhopal, India

Professional Summary

Third-year B.Tech student at IIIT Bhopal (CPI 8.49/10) specializing in **production agentic AI systems** – designing, engineering, and shipping multi-agent pipelines that run reliably in production. Strong CS fundamentals (DSA, OOP, systems) reinforced by active competitive programming (**CodeChef 4-Star**, **Codeforces Pupil**). Currently ML Intern at Moleculyst, engineering multimodal inference pipelines with structured evaluation loops, observability, and reproducible deployment. Shipped end-to-end production systems including a **5-agent LangGraph orchestration** on Google Cloud Run (Equitas AI), a **multimodal agentic VLM workspace with automated critique loops** (Moleculyst Studio), and a **containerized reinforcement learning environment with strict API contracts** (Triage AI). Comfortable across the stack – FastAPI, Docker, Pydantic, Vertex AI – and committed to availability, observability, and clean engineering at scale. Excited to build agentic AI infrastructure for ambitious drug-discovery research.

Education

IIIT Bhopal Bhopal, Madhya Pradesh, India
B.Tech in Information Technology (CS & Engineering Curriculum) | CPI: 8.49 / 10 Aug 2024 – May 2028

- **Relevant Coursework**: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks, Discrete Mathematics, Linear Algebra, Probability & Statistics

Technical Skills

Languages: Python, C++, C, SQL

Software Engineering: End-to-End Software Design, REST API Development (FastAPI), Software Testing & Debugging, Version Control (Git/GitHub), API Contracts (Pydantic), On-Schedule Delivery

Agentic AI / Orchestration: LangChain, LangGraph, Multi-Agent Orchestration, Tool Use, Automated Critique Loops, OpenEnv, Reinforcement Learning Environments

LLM / VLM Stack: LLaMA 3.1, Gemini API, VLMs, Multimodal LLMs, Prompt Engineering, RAG

ML / Deep Learning: PyTorch, TensorFlow, Scikit-learn, LSTM, XGBoost, SHAP

Infrastructure & Deployment: Docker, AWS, GCP (Cloud Run, Vertex AI), Scalable Pipeline Architecture, Availability, Reliability, Observability, Monitoring

CS Fundamentals: Data Structures, Algorithms, Object-Oriented Design, Time/Space Complexity Analysis

Competitive Programming (C++): CodeChef 4-Star (max 1856), Codeforces Pupil (max 1340) – DP, graphs, segment trees, combinatorics

Experience

ML Intern Feb 2026 – Present
Moleculyst Remote

- Engineered **multimodal Video LLM inference pipelines** in Python – integrating vision-language architectures, preprocessing stages, and structured evaluation loops – delivering Moleculyst's core inference system for structured understanding across complex operational video datasets.
- Improved pipeline **reliability and efficiency** by designing structured evaluation frameworks that standardized quality assessment across model versions, enabling reproducible benchmarking without manual intervention.
- Adopted new engineering methods across multimodal AI and applied quantitative research – reviewing current developments and incorporating probability theory, mathematical optimization, and statistical modeling to improve production system performance.
- Collaborated with cross-functional stakeholders to determine user requirements, iterate on technical designs, and apply best practices – **delivering software projects on schedule** in a cooperative team environment.

Projects

Equitas AI – 5-Agent Production Pipeline on Google Cloud *Python, LangGraph, FastAPI, Gemini API, Vertex AI, GCP* 2025

- Designed a **5-agent autonomous LangGraph pipeline** for dataset bias auditing, orchestrating iterative mitigation loops that converged at Disparate Impact Ratio (DIR) ≥ 0.80 across all evaluated datasets.
- Deployed a **keyless FastAPI backend on Google Cloud Run via Vertex AI** – fully serverless production system with autoscaling availability, no stored secrets, and zero credential management overhead.
- Integrated SHAP explainability matrices with Gemini-driven counterfactual narration through an interactive review interface usable by non-technical stakeholders.

Moleculyst Studio – Multimodal Agentic Workspace with Observability *Python, Gemini API, VLMs, OpenCV, Docker* 2025

- Built a multi-step **agentic editing pipeline** composing VLM-based grounding, targeted edit operators, and an **automated critique loop** – replacing manual step-by-step review with a single-command end-to-end workflow.
- Engineered **full pipeline observability** by streaming structured reasoning traces, tool calls, and per-step quality scores into a unified monitoring view – enabling engineers to inspect, debug, and intervene at any stage.

Triage AI – Containerized Reinforcement Learning Environment *Python, FastAPI, OpenEnv, Pydantic, Docker, LLaMA 3.1* 2025

- Architected a **reinforcement learning environment** for automated email classification by implementing structured OpenEnv state-machine cycles (reset / step / observe) with multi-level prompt-engineering chains, achieving 0.71 / 1.0 reward accuracy on LLaMA 3.1 (8B).
- **Containerized the full system with Docker** and **enforced API contracts with Pydantic** – reproducible, environment-isolated deployment that can be reliably reset and re-evaluated across test cycles.

PharmAI – Production Forecasting System *Python, LSTM, XGBoost, SHAP, Scikit-learn* 2025

- Engineered an LSTM + XGBoost ensemble for pharmaceutical batch quality prediction, achieving >**90% accuracy** on held-out validation batches, with SHAP-driven anomaly attribution for interpretable corrective action.

Achievements

- **Meta × PyTorch OpenEnv Hackathon – Grand Finalist** (Bangalore, 2025): Top-tier ranking out of 52,000+ global participants for shipping a production-ready reinforcement learning environment.
- **CodeBidz Hackathon – 1st Place**: Won as a solo developer, independently executing the full product lifecycle – architecture, engineering, and delivery – under competition time constraints.
- **OpenAI × Outskill AI Builders Hackathon**: Shortlisted in Top 1,000 developers nationwide to design, optimize, and ship a production-grade live AI product.
- **JEE Main 2024 – 98.16 percentile**: Top 1% nationally out of 1.2M+ candidates; qualified for JEE Advanced.
- **SRMC Mathematics Competition – AIR 27 (Finals)**: National finalist in the tournament jointly organized by IIT Bombay & IIT Madras.